



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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FRAUD TRANSACTION ALERTING SYSTEM

Subject : Python programming
Subject code : CSA0808

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ABSTRACT



- Designed a Python-based system to detect and prevent fraudulent financial transactions.
- Digital transactions are growing rapidly, increasing the risk of fraud.
- This project aims to detect fraudulent transactions using simple monitoring techniques.
- It analyzes user behavior and verifies transactions using authentication and OTP.
- The system helps prevent unauthorized access and improves transaction security.



Objectives



- **Real-Time Transaction Monitoring** Monitor every financial transaction instantly.
- **Fraud Detection** Identify suspicious transactions based on amount, location, time, and user behavior.
- **Risk Classification** Categorize transactions as **Normal**, **Suspicious**, or **Fraudulent**.
- **Instant Alert Generation** Notify users immediately when suspicious activity is detected.
- **OTP Authentication** Verify high-risk transactions using a secure One-Time Password (OTP).



INTRODUCTION



- Digital transactions are increasing rapidly in today's world
- This also increases the risk of fraud and unauthorized activities
- Fraud detection systems help identify suspicious transactions
- They analyze patterns like amount, location, and user behavior
- Alerts or OTP verification are triggered for risky transactions
- Ensures security and protects users from financial loss

LITERATURE REVIEW

Author	Year	Key Ideas	Methodology
Zhang et al.	2023	AI-based fraud detection in digital payment systems	Deep Learning (LSTM & Neural Networks)
Sharma & Verma	2023	UPI and online banking fraud prevention	Machine Learning with transaction pattern analysis
Kim et al.	2024	Real-time fraud detection using Artificial Intelligence	Random Forest and XGBoost algorithms
Patel & Rao	2025	Fraud prediction using behavioral analytics	User behavior analysis with anomaly detection
Ahmed et al.	2026	Intelligent fraud transaction alerting system	Hybrid AI model with real-time alerts and OTP verification



Problem statement



Problem Statement

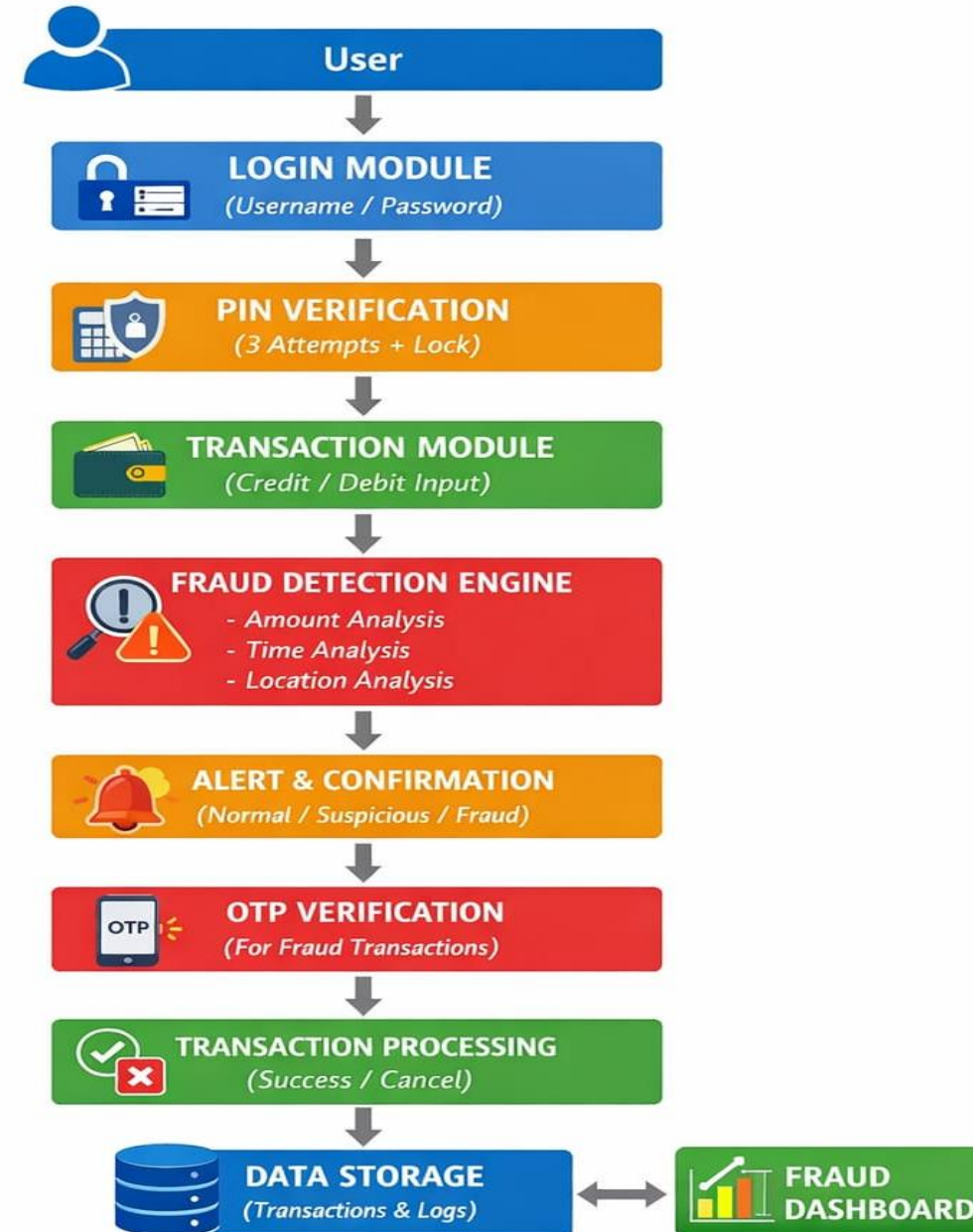
Financial fraud has become a major challenge with the rapid growth of digital banking, online payments, and e-commerce. Traditional transaction monitoring systems often fail to detect suspicious activities in real time

- Allowing unauthorized transactions to occur and causing significant financial losses.
- Many existing systems also lack intelligent analysis of user behavior and effective multi-layer security mechanisms.

FRAUD DETECTION PROCESS:

Fraud Detection System

System Architecture Diagram





MODULE 1: TRANSACTION CHECK



- This module collects transaction details entered by the user such as amount, type, location, and category.
- It supports both credit and debit transactions to simulate real banking operations.
- The system records each transaction in real-time for further processing and analysis.
- It maintains a history of transactions which helps in identifying user behavior patterns.
- The module ensures that all input data is properly structured before sending it to the detection system.



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Amount

Type
debit

Category
food

Location
Chennai

Submit



MODULE 2: FRAUD DETECTION SYSTEM



- This is the core module that analyzes each transaction to identify possible fraudulent activities.
- It calculates a fraud score based on factors like transaction amount, time, and location changes.
- The system detects unusual patterns by comparing current transactions with past user behavior.
- It classifies transactions into normal, suspicious, or fraudulent categories.
- The module works in real-time to ensure immediate detection of risky transactions.



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MODULE 3: ALERT AND OTP MODULE

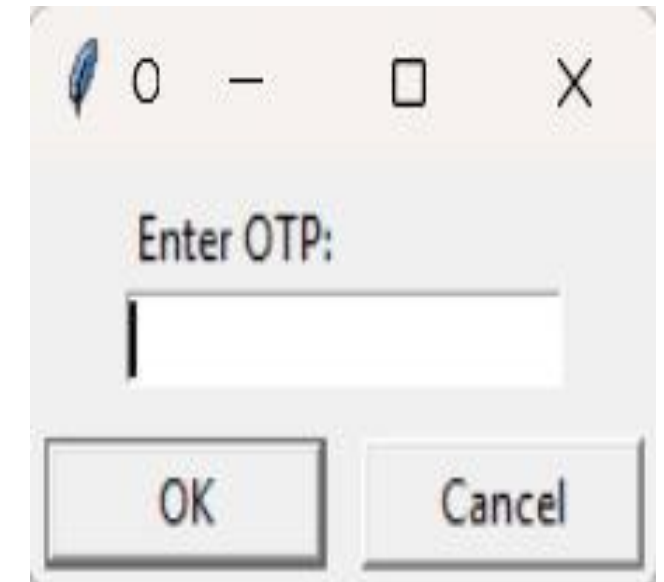
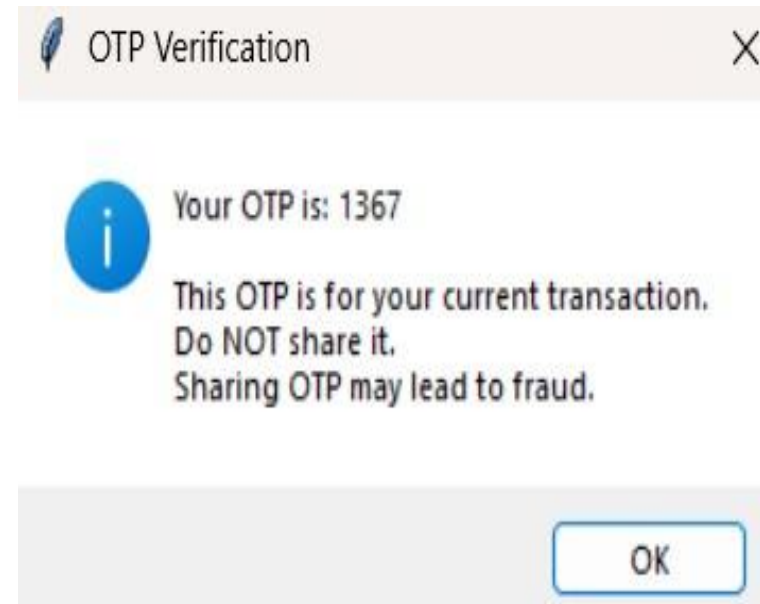
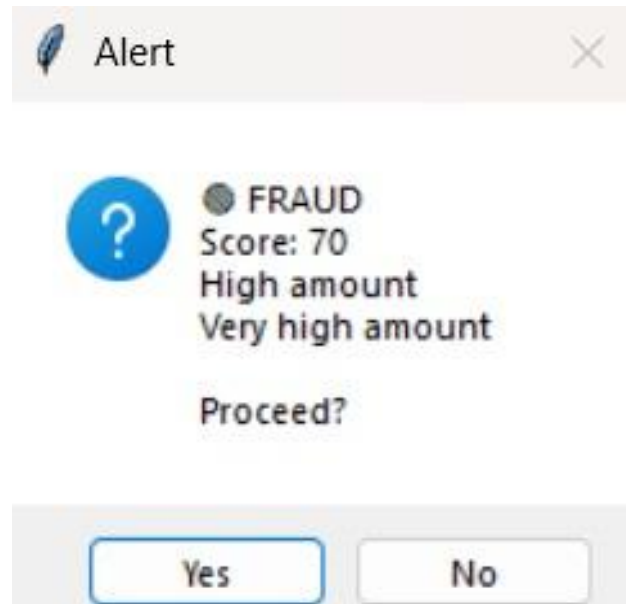


- This module generates alerts whenever a transaction is detected as suspicious or fraudulent.
- It asks the user for confirmation before allowing the transaction to proceed.
- An OTP (One-Time Password) is generated for secure verification of the transaction.
- The user must enter the correct OTP to successfully complete the transaction.
- If the OTP is incorrect, the transaction is immediately cancelled to prevent misuse.
- This module adds an extra layer of security to protect user accounts from unauthorized access.



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INTEGRATION



- The Transaction Module collects transaction details like amount, type, and location from the user.
- This data is immediately sent to the Fraud Detection Module for analysis. The Fraud Detection Module checks for unusual patterns and calculates the fraud risk.
- If the transaction is normal, it is processed without interruption.
- If the transaction is suspicious or fraudulent, it is passed to the Alert & OTP Module.
- The Alert & OTP Module sends an alert and asks for OTP verification before completing the transaction.

References:

- Zheng ,“ Transaction fraud detection based on total order relation and behavior diversity." IEEE Transactions on Computational Social Systems 5.3 (2023): 796-806.
- L., Liu, G., Transaction fraud detection based on total order relation and behavior diversity. IEEE Transactions on Computational Social Systems, 5(3), 796-806.
- Guanjun Liu,. "Transaction fraud detection based on total order relation and behavior diversity." IEEE Transactions on Computational Social Systems 5, no. 3 (2022): 796-806.
- Jiang, C., 2023. Transaction fraud detection based on total order relation and behavior diversity. IEEE Transactions on Computational Social Systems, 5(3), pp.796-806.

Conclusion

- The Fraud Transaction Detection System successfully identifies suspicious and fraudulent activities using real-time analysis.
- It enhances security through multi-level authentication, including PIN verification and OTP validation.
- The system helps in preventing unauthorized transactions and protecting user data.
- With features like alert mechanisms and account locking, it improves reliability and user trust.
- This project can be further extended and implemented in real-world banking systems for advanced security and fraud prevention.

THANK YOU